Introspective Imitation Dynamics and the Ceiling of Cooperation in Extrinsic Update Rules

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June 5, 2026

Abstract

Evolutionary game theory provides a framework by which to study the emergence of cooperation in a population of self-interested actors. In such a framework, players' decisions on whether or not to cooperate evolve according to decision rules called population dynamics. However, often games are studied under the assumption that all individuals play under the same conditions. As such, many common choices of update rule are not well suited for a heterogeneous population. In this paper, we categorise and compare four different population dynamics as "extrinsic", where players learn by looking outward at the other players in the population, and "intrinsic", where players look inwardly at their own attributes or potential payoffs. We show that extrinsic population dynamics have a ceiling on the rate of cooperation which can be exceeded by intrinsic population dynamics, and demonstrate this using the public goods game with heterogeneous contributions. We then introduce "introspective imitation dynamics", a novel population dynamic which combines both fitness-proportional extrinsic selection and self-referential intrinsic update steps, which overcomes the inherent limitations of the other four population dynamics.

1 Introduction

Reducing global carbon emissions is one of the defining collective action problems of our time. Countries differ substantially in their economic capacity, historical emissions, and the costs they face when decarbonising, yet they share a mutual interest in the global public good of a stable climate. However, they face the temptation to free-ride, accepting the benefits of the public good without contributing to the cost[40]. International agreements have attempted to encourage a reduction in carbon emissions, for example the Kyoto protocol[1] and Paris agreement[7, 26]. A recent example is the European Union's Carbon Border Adjustment Mechanism [13], which requires importers to purchase certificates reflecting the carbon content of goods produced under weaker environmental regulation, represents a real-world instance of this idea.

This structure maps onto a *public goods game* [4, 23, 25, 24, 51]: each actor decides whether to incur a private cost to contribute to a shared resource whose total value is multiplied by some factor r and distributed equally amongst all players. The values $1 < r < N$ are usually considered; the case where $r > N$ is not as well studied. This creates the temptation to free-ride, and so purely selfish reasoning leads to a tragedy of the commons [22, 61], even when collective cooperation would benefit everyone. Both theory [49] and controlled experiments [38, 37] show that framing cooperation as a collective-risk problem, where failure to meet a cooperation threshold incurs a shared loss, can substantially raise cooperation rates. The challenge of overcoming free-riding is not merely theoretical. Nordhaus [40] proposed voluntary *climate clubs* in which member nations adopt a common carbon price and levy tariffs on non-members, turning the international climate problem into a coordination game with enforcement.

Evolutionary game theory offers a framework for studying how cooperation can emerge among self-interested actors without assuming rational foresight [27, 41]. In this approach, strategies that perform well spread through a population via one of several update rules, while poorly performing strategies decline. The conditions under which cooperative strategies can invade and stabilise have been extensively studied in two-player settings such as the iterated prisoner's dilemma [31, 6] and the snowdrift game [60]. The public goods game generalises these pairwise dilemmas to N players [23], making it well suited to studying multi-lateral cooperation such as climate agreements. In finite populations, evolutionary outcomes are stochastic: a rare mutant strategy spreads or vanishes with a probability that depends on the population size, payoff structure, and update rule [43, 57].

A feature of real-world collective action that standard models typically set aside is that actors are not identical. Players differ in wealth, productive capacity, and the costs they face, and this inequality can reshape the dynamics of cooperation [25]. In spatial and graphical settings, where players sit on a network and participate simultaneously in multiple games with their neighbours [42, 45, 54], heterogeneity generally promotes cooperation. In [50] it is shown that social diversity in endowments promotes the emergence of cooperation in the public goods game directly: heterogeneous populations outperform their homogeneous counterparts over a wide range of parameters. This echoes earlier work showing that scale-free network topology, a form of structural heterogeneity, provides a unifying framework for the emergence of cooperation [48]. When contributions depend on dynamically accumulated reputation [36], high-reputation defectors receive reduced transfers, weakening defection's advantage. When heterogeneity arises from differences in network degree so that players allocate contributions according to the strength of their social ties [32], the payoff structure penalises high-degree defectors and protects cooperating clusters from invasion. The characteristic spatial mechanism is the formation of cooperating clusters that are more profitable than their defecting neighbours; provided r is sufficiently large, these clusters resist invasion and eventually spread. Heterogeneity does not, however, unconditionally favour cooperation. Flores et al. [14] show that when the contribution level is itself a strategic variable rather than a fixed attribute, low-value contributions act as a form of defection against the most collectively beneficial strategies, so that the emergence of cooperation does not guarantee the emergence of high-value cooperation.

The conditions under which one strategy may invade a population has been studied in depth for certain evolutionary processes. The $\frac{1}{N}$ fixation probability for a single mutant in neutral drift [41] has been used as a benchmark for when a strategy is favoured by selection. Results such as the $\frac{1}{3}$ rule [44, 8] have been shown to hold even when a population updates with both the Moran process and Fermi imitation dynamics [34]. Much study also focuses on symetric 2x2 games with payoff matrix:

	A	B
A	a	b
B	c	d

It has been shown that the condition $a(N-2)+bN > cN+d(N-2)$ [29] in such games leads to an abundance of strategy A. This has been shown to hold for a wide range of evolutionary processes [3]. One strategy may dominate strictly under certain relationships between $a, b, c,$ and d [2]. The effects of mutation on such a game have also been studied [3, 19], with the latter showing that under action-invariant mutation, $b > c, a > d$ gives domination for A players. However, these results are given for a symetric game played pairwise between members of the population, rather than a heterogeneous population. We extend this work to consider all types of games where a defector outperforms a cooperator in a given population.

How the choice of update rule interacts with population heterogeneity is also poorly understood. The Moran process [30, 39], Fermi imitation dynamics [55], aspiration dynamics [12], and introspection dynamics [9, 10] all model strategy revision in distinct ways, and each may respond differently to player-specific payoffs. In particular, *intrinsic* dynamics, in which a player evaluates a prospective strategy change based on their payoff in comparison with their own counterfactual payoff or aspiration level, are better suited to heterogeneous populations than *extrinsic* dynamics, in which a player may copy a high-fitness neighbour without considering whether the same strategy would serve them as well.

There is also precident in the literature for creating population dynamics from the combination of other processes. Previous works have studied this in terms of evidential reasoning [46], populations where the population dynamic is a heterogeneous attribute of the players [35, 5, 33], players choosing an update rule probablistically [64, 5, 59], and explicit combinations of steps from different population dynamics [58]. Many of these works have focused on the combination of aspiration and imitation learning rules, or the combination of different methods of social learning, for instance combining the Moran process with Fermi imitation dynamics. The former method is of particular interest, as it combines social learning with aspirational self-asessment, creating a process in which a player considers multiple measures of a strategy's success when updating their action type.

We make the following contributions. First, we build on an established general framework for evolutionary dynamics on fully heterogeneous, well-mixed populations of N ordered individuals, in which each player may have a distinct payoff function and contribution level. Second, we formalize the notion of a purely extrinsic population dynamic, a definition encompassing both the Moran process [30, 39] and Fermi imitation dynamics [55], and show that such dynamics are inherently constrained by an upper bound on the probability of cooperation. Third, we consider two purely intrinsic population dynamics, aspiration dynamics [12] and introspection dynamics [9, 10], showing that these processes can overcome the ceiling of purely extrinsic dynamics, and discuss their weaknesses. Finally, we introduce a novel update rule, *Introspective Imitation Dynamics*, in which players select a role model proportionally to fitness (as in the Moran process) but evaluate whether to adopt that strategy by comparing their own current and counterfactual payoffs (as in introspection dynamics). This rule is well suited to heterogeneous populations, where copying a high-fitness individual occurs with a high probability, but is irrational if the considered strategy yields a worse payoff for the copying player. For each population dynamic, we will use the public goods game [51, 25, **?**, 24, 23] to demonstrate our results.

The remainder of the paper is organised as follows. Section 2 introduces the population model and the public goods game. Section 3 introduces a formal definition of a *purely extrinsic population dynamic*, gives formulations of the Moran process and Fermi imitation dynamics in terms of our population model, proves that these satisfy the definition of a purely extrinsic population dynamic, and proves the ceiling on the probability of cooperation in such dynamics. We also demonstrate this in a public goods game. Section 4 gives formulations of two intrinsic population dynamics and shows that they can exceed the ceiling on the probability of cooperation. We also discuss their disadvantages, and show their results in the public goods game. Section 5 introduces *introspective imitation dynamics*, a novel population dynamic, and show that it exhibits none of the disadvantages present in the other four dynamics. We then show under which conditions introspective imitation dynamics is able to exceed the cooperation ceiling in the public goods game.

2 The General Heterogeneous Population Dynamic Model

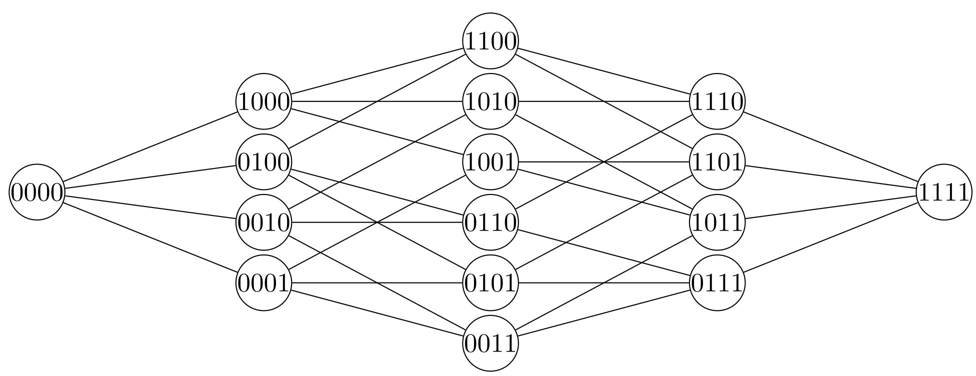
As in [9, 10], we define a general framework for evolutionary dynamics on a fully heterogeneous, well-mixed population.

2.1 The State Space

We model a population consisting of N **ordered** individuals. Each individual selects an action from a common action set $A = (a_1, a_2, \ldots, a_k)$; we assume throughout that all individuals share the same action set of size k. A *state* of the population is an ordered N-tuple $\mathbf{a} = (a_1, a_2, \ldots, a_N) \in A^N$, and the *state space* S is the set of all such tuples. A payoff function $\pi\colon S \to \mathbb{R}^N$ assigns to each state a vector of individual payoffs; we write $\pi_i(\mathbf{a})$ for the payoff of individual i in state $\mathbf{a}$.

The state space of an evolutionary game can be described as the graph who's nodes are the states $\mathbf{a} \in S$, with an edge between two vertices $\mathbf{a}, \mathbf{b}$ if and only if they differ in exactly one position. This defines an evolutionary process in which one player changes action type at each time step. Each state also has a self-loop, as at a given time step, a player may accept the same strategy they currently posess, or reject the strategy which they have considered. We call the unique coordinate at which $\mathbf{a}$ and $\mathbf{b}$ differ the *index of difference*, denoted $I(\mathbf{a}, \mathbf{b})$ [10]. The set of states reachable from $\mathbf{a}$ in one step is denoted $\mathrm{Neb}(\mathbf{a}) = \{\mathbf{b} \in S : h(\mathbf{a}, \mathbf{b}) = 1\}$. This notation follows [10].

In the case of a game with two actions, the state space of the Markov chain is analogous to an N-dimensional hypercube. This is the graph whose node set consists of the 2^N N-dimensional boolean vectors, and which has edges between edges if and only if they differ in exactly one position [21]. The state space of an evolutionary game with two actions can be modelled in this way by denoting the strategies 0 and 1. The graphical representation of a birth-death process differs from a hypercube in exactly two ways. First, a population may remain the same at a given step, and thus each vertex contains a self-loop. Second, the probability of transitioning from a state $\mathbf{a}$ to a state $\mathbf{b}$ is not necessarily the same as the probability of transitioning from $\mathbf{b}$ to $\mathbf{a}$. Thus, instead of one edge between two states, we have two directed edges, $\mathbf{a} \to \mathbf{b}$ and $\mathbf{b} \to \mathbf{a}$.Figure 1 shows the 4 dimensional hypercube.



(a)

Figure 1: The hypercube of dimension 4. This is closely related to the graph of a 4-player population dynamic.

2.2 Population Dynamics and mutation

The population evolves as a Markov chain [52] on S. We therefore obtain a transition matrix T, where $T_{\mathbf{ab}}$ denotes the probability of transitioning from a state $\mathbf{a}$ to a state $\mathbf{b}$. They take the general form:

$$T_{\mathbf{ab}} = \begin{cases} P(\mathbf{a} \to \mathbf{b}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{1}$$

where $P(\mathbf{a} \to \mathbf{b})$ is the probability that player $I(\mathbf{a}, \mathbf{b})$ changes from its action type in $\mathbf{a}$ to its action type in $\mathbf{b}$. The process which determines this probability is called a *population dynamic*. We shall discuss five population dynamics in this paper, and classify them as shown in Figure 2. Purely extrinsic population dynamics only consider the success of a strategy in the current population when updating a player's action type. This models a behaviour where players look outwardly to attempt to improve their payoff. Purely intrinsic population dynamics model a player who looks inwardly to improve their payoff - comparing their current payoff to some value which does not depend on the payoffs of an action type in the current population. The final classification is our novel population dynamic, introspective imitation dynamics, which contains both an extrinsic step and an intrinsic step.

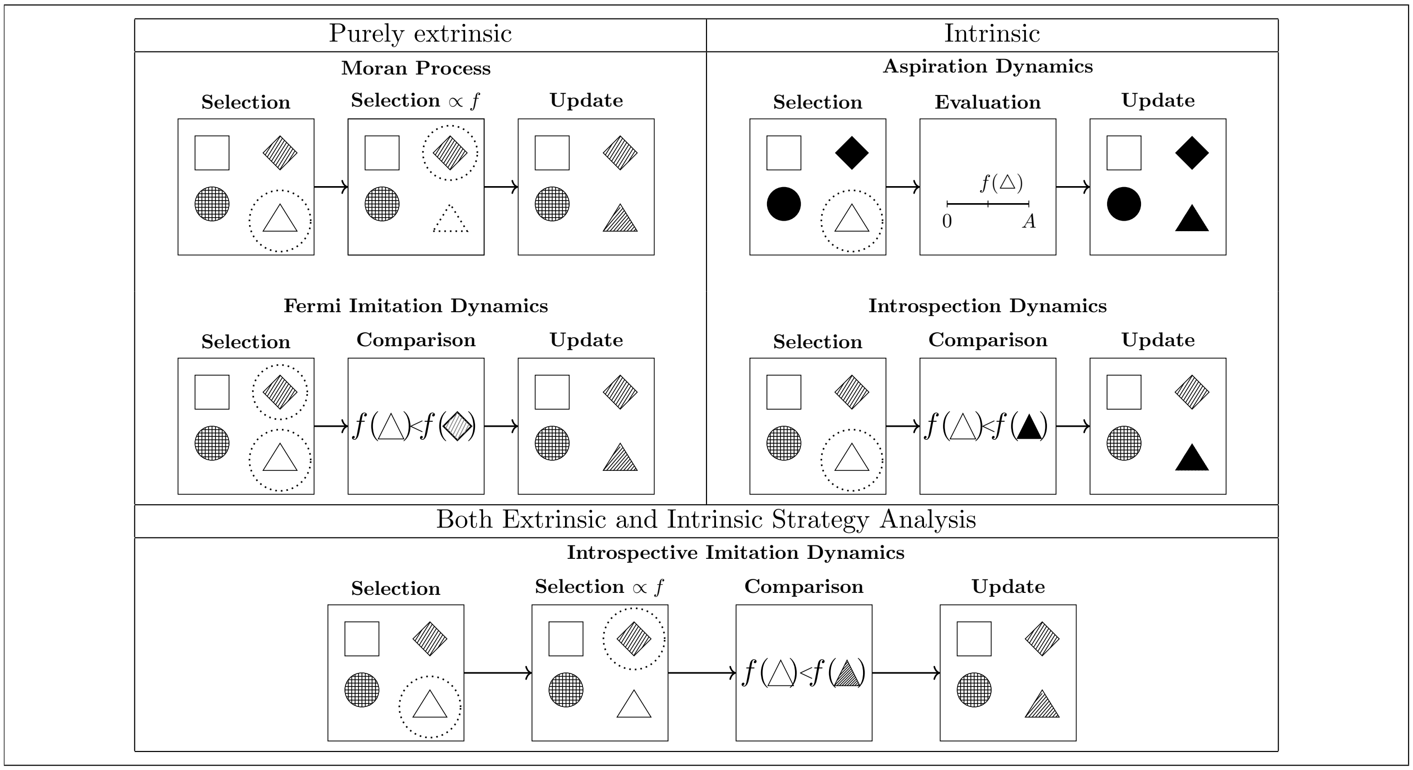


Table 2: A comparison of different population dynamics. We assume in all cases that the triangle is chosen for replacement with a probability $\frac{1}{N}$ and the diamond is chosen with a varying probability depending on the process. In the case of introspection dynamics, we assume that the strategy represented by the shape being filled in is chosen with probability $\frac{1}{k-1}$, and for aspiration dynamics there is only one other strategy to accept.

Three of the dynamics discussed in this paper correspond to absorbing Markov chains, where the measurable outcome is the probability of absorption into a given absorbing state. The remaining two dynamics are ergodic chains, for which we can compute the long-run abundance of a given strategy in the population. To enable meaningful comparison across all models, we introduce mutation. Mutation ensures that all processes become ergodic, while also introducing a controlled amount of noise: at any moment, individuals may switch to a different strategy [19].

Given a process defined by transition matrix T and two states $\mathbf{a}$ and $\mathbf{b}$, the probability of the original process *with mutation* transitioning from $\mathbf{a}$ to $\mathbf{b}$ is given by:

$$P^{(\mu)}(\mathbf{a} \to \mathbf{b}) = P(\text{pick individual } I(\mathbf{a}, \mathbf{b})) \cdot \big(P(\mathbf{a}_{I(\mathbf{a},\mathbf{b})} transitions \to \mathbf{b}_{I(\mathbf{a},\mathbf{b})}) \cdot P(\text{no mutation}) + P(\mathbf{a}_{I(\mathbf{a},\mathbf{b})} \text{ mutates} \to \mathbf{b}_{I(\mathbf{a},\mathbf{b})})\big)$$

We assume a mutation parameter $\mu \in [0,1]_{\mathbb{R}}^{N \times k}$, where μ_{ij} denotes the probability that individual i adopts strategy j via mutation. For a focal individual i, the probability that no mutation occurs is $1 - \sum_{j=1}^{k} \mu_{ij}$. Let T be the transition matrix for a population dynamic and $T_{\mathbf{ab}}$ be the transition probability from a state $\mathbf{a}$ to a state $\mathbf{b}$. Then the transition matrix $T^{(\mu)}$ for the process with mutation is given by:

$$T^{(\mu)}_{\mathbf{ab}} = \begin{cases} T_{\mathbf{ab}}\left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{2}$$

Note that if $\sum_{j=1}^{k} \mu_{ij} = 1$, mutation fully determines the behavior of individual i. This construction ensures that all processes are ergodic, rather than absorbing. Consequently, we no longer obtain absorption matrices for any model with mutation; instead, we derive a unique steady state that is independent of the initial state. All processes in this paper are considered under the effect of action-invariant mutation, where $\mu_{iw} = \mu_{il}$ for all action types w, l. This ensures that mutation does not cause an inherent bias towards a given action type.

3 Extrinsic population dynamics

We begin by defining a *purely extrinsic population dynamic*. Intuitively, a purely extrinsic process revises strategies through payoff comparison between members of a population.

Definition 1 (Purely extrinsic population dynamic)**.** *Let T be the transition matrix of a population dynamic on a state space S. We say that T defines a* purely extrinsic *process if for every $\mathbf{a}, \mathbf{b} \in S$ with $\mathbf{b} \in Neb(\mathbf{a})$, $T_{\mathbf{ab}}$ satisfies:*

1. *$T_{\mathbf{ab}}$ is a function of payoffs evaluated in $\mathbf{a}$ and in no other state.*
2. *Let $\mathcal{R}$ denote the set of players j such that $j \neq I(\mathbf{a},\mathbf{b})$ and $\mathbf{a}_j = \mathbf{b}_{I(\mathbf{a},\mathbf{b})}$. Then $T_{\mathbf{ab}}$ is non-decreasing in the payoffs $\pi_j(\mathbf{a})$ for all $j \in \mathcal{R}$ (and strictly increasing in at least one such j if $\mathcal{R} \neq \emptyset$), and non-increasing in the payoffs $\pi_j(\mathbf{a})$ for all $j \notin \mathcal{R}$.*
3. *$T_{\mathbf{ab}}$ contains no term dependent on $a_{I(\mathbf{a},\mathbf{b})}$ or $b_{I(\mathbf{a},\mathbf{b})}$, except in the payoff function $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$, mutation parameters μ_{ij}, and in the definition of the set $\mathcal{R}$.*

Condition (2) captures the basic monotonicity: a higher payoff for a role model makes that role model more likely to be copied, and a higher payoff for a non-role-model pulls the other way. Condition (3) rules out an inherant bias towards either action. Without (3), we must consider population dynamics which explicitly favour a certain action type, for example:

$$\bar{T}_{\mathbf{ab}} = \begin{cases} \delta_{I(\mathbf{a},\mathbf{b})} \cdot T_{\mathbf{ab}} & \text{if } \mathbf{b} \neq \mathbf{a}, \\ 1 - \sum_{\mathbf{c} \neq \mathbf{a}} T_{\mathbf{ac}} & \text{if } \mathbf{b} = \mathbf{a}. \end{cases}$$

where $T_{\mathbf{ab}}$ is the transition matrix of some purely extrinsic dynamic, and $\delta_{I(\mathbf{a},\mathbf{b})}$ takes the value 1 if $b_{I(\mathbf{a},\mathbf{b})} = C$ and $\frac{1}{2}$ if not. Without (3), $\bar{T}$ would define purely extrinsic population dynamic, but would be tailored to favour a particular action type. We exclude cases such as this, as they do not conform to the framework in which fitness is the single measure of the success of action types in the population, and thus of which strategy is passed on in each step of the process. While parameters such as choice intensity and selection intensity affect the transition probability $T_{\mathbf{ab}}$, they do not inherently favour any specific action type.

We now show two classical examples of purely extrinsic population dynamics

3.1 The Moran Process

2a

The Moran process [39] is defined by the following algorithm:

1. A player is chosen to be copied with a probability proportional to their fitness in the population
2. A player is chosen with probability $\frac{1}{N}$ to change their strategy

And the transition matrix is given by:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{\sum_{j:\, a_j = b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{N \sum_j f_j(\mathbf{a})} \left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{3}$$

Where $f_i(\mathbf{a}) = 1 + \epsilon \pi_i(\mathbf{a})$ is the fitness of a player i in state $\mathbf{a}$ scaled with selection intensity ϵ. This ensures a positive fitness for all players in order that the probability $T_{\mathbf{ab}}$ is always well defined. The selection intensity ϵ controls the rationality with which players choose which strategy to adopt in each time step - a larger ϵ indicates that a player will choose a higher fitness strategy with a greater probability. This value of ϵ must, however, be chosen in order that $f_i(\mathbf{a})$ is strictly positive.

The Moran process is traditionally employed as a model of natural evolution [39, 41], where a player's "type" represents a species or allele within a population. In each time step, rather than a given player choosing a new action type, a randomly chosen individual in the population would die and be removed from the population. At the same time, an individual would reproduce with a probability proportional to their fitness, introducing an exact copy of themselves to the population to replace the removed individual. As a result of the model's original purpose, the Moran process contains no decision step where a player chooses whether or not to accept a considered strategy. Instead, at each time step, a player *must* accept another player's action, and we remain in the same state when a player accepts the same action as they currently play.

One implication of this is that in a heterogeneous population, worsening moves may be taken with a high probability, as an action that provides a high payoff for one player in a population may not provide the same return for another. In nature, an offspring is unable to choose its parents, and thus cannot choose which attributes it is born with. Furthermore, parents will choose a partner based on its success in the population, and thus traits which give a high fitness will be passed on with high probabilities, regardless of how they will perform for the offspring. When modelling the adoption of behaviours, these assumptions may lead to players making decisions which are harmful for their own payoff. For example, in the public goods game with heterogeneous returns, a player with a low $r_i < N$ may copy a player who contributes with a high $r_j > N$ due to their high individual payoff. However, this would be a worsening move for the focal player i, as due to their low individual rate of return they would lower their payoff.

We now show that the Moran process satisfies the conditions of Definition 1

Theorem 1 (The Moran process is a purely extrinsic process)**.** *The Moran process, as defined by Equation* (3)*, is a purely extrinsic population dynamic.*

Proof. By definition the Moran process satisfies conditions (1) and (3). It remains to be shown that $T_{\mathbf{ab}}$ is non-decreasing in $\pi_j(\mathbf{a})$ for $j \in \mathcal{R}$ (strictly increasing for at least one such j) and non-increasing in $\pi_j(\mathbf{a})$ for $j \notin \mathcal{R}$. Recall $f_i(\mathbf{a}) = 1 + \epsilon \pi_i(\mathbf{a})$ for some selection intensity $\epsilon > 0$. Then we have:

$$T_{\mathbf{ab}} = \frac{\sum_{j \in \mathcal{R}} f_j(\mathbf{a})}{N \sum_{l=1}^{N} f_l(\mathbf{a})} \left(1 - \sum_{i=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),i}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$$

$$\frac{dT_{\mathbf{ab}}}{d\pi_j(\mathbf{a})} = \frac{\epsilon \sum_{x \notin \mathcal{R}} f_x(\mathbf{a})}{N \left(\sum_{l=1}^{N} f_l(\mathbf{a})\right)^2} \left(1 - \sum_{i=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),i}\right) \quad \text{for } j \in \mathcal{R}$$

For $j \in \mathcal{R}$, as ϵ is chosen such that $f_i(\mathbf{a}) > 0$ for all $i \in \mathbf{a}$, and $\mu_{i,j}$ such that $\sum_{i=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),i} < 1$, this derivative is strictly positive. For $j \notin \mathcal{R}$, the analogous computation gives $\frac{dT_{\mathbf{ab}}}{d\pi_j(\mathbf{a})} = -\frac{\epsilon \sum_{x \in \mathcal{R}} f_x(\mathbf{a})}{N(\sum_{l=1}^{N} f_l(\mathbf{a}))^2}(1 - \sum_{i=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),i}) \leq 0$, so condition (2) holds. $\square$

3.2 Fermi Imitation Dynamics

2c

Fermi Imitation Dynamics [55], or simply imitation dynamics, is another method of modelling evolution in a population. It follows the process:

- Two players, i and j are selected at random. Unlike the original formulation, player j need not be a neighbour of player i, reflecting the well-mixed setting considered here.

- Player i copies the action type of the player j with a probability according to the Fermi function $\phi(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a})) = \frac{1}{1+e^{(\beta(\pi_i(\mathbf{a})-\pi_j(\mathbf{a})))}}$

Where β is the *choice* intensity of the system, representing how sensitive our players' choice is to the difference between their current payoff, and the payoff which they are considering. This method has been commonly used in studies of evolutionary games, for example [36, 32, 55]. We now define our transition matrix according to the Fermi function:

$$T_{\mathbf{ab}} = \begin{cases} \frac{1}{N(N-1)} \sum_{j:\, a_j = b_{I(\mathbf{a},\mathbf{b})}} \phi\left(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_j(\mathbf{a})\right) \left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{4}$$

Unlike a Moran Process, we can see that the players are not selected with a probability proportional to their fitness. Instead, Fermi Imitation Dynamics selects a pair of players and compares their payoffs directly, giving players the option not to accept a strategy in a given time step. However, similarly to the Moran process, in a heterogeneous population the method of adopting a new strategy based entirely on the payoff of another individual may lead to worsening moves in which players adopt strategies which currently admit a higher fitness, but which will not improve their own fitness.

We also see that due to the formulation of the logit function $\phi(\Delta(\pi))$, Fermi imitation dynamics can become invariant under certain parameters. This occurs in any game in which one action type's payoff in any given state can be expressed as another action type's payoff up to the addition of some factor γ, where γ depends only on a subset of the game's parameters. In that case, $\Delta(\pi) = \pm\gamma$, and is therefore invariant to any parameters of the game which do not influence γ. Take, for example, the public goods game, with a player's payoff $\pi_i(\mathbf{a}) = \frac{r}{N} \sum_{j=1}^{N} \alpha_j - \alpha_i$ for contribution α_i and rate of return r (here $\alpha_i = 0$ if i defects). When comparing a cooperator and defector under ϕ, we have the value $\Delta(\pi) = \pm\alpha_i$, and thus we see that the parameter r has no effect on the probability of accepting a new strategy under this population dynamic.

We now show that Fermi imitation dynamics satisfies the definition of a purely extrinsic population dynamic.

Theorem 2 (Fermi imitation dynamics is a purely extrinsic population dynamic)**.** *Fermi imitation dynamics, as defined by Equation* (4)*, is a purely extrinsic population dynamic.*

Proof. By definition, Fermi imitation dynamics satisfies conditions (1) and (3). It remains to show that it satisfies condition (2).

$$T_{\mathbf{ab}} = \frac{1}{N(N-1)} \sum_{j \in \mathcal{R}} \phi(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_j(\mathbf{a})) \left(1 - \sum_{i=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),i}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$$

As ϕ is strictly decreasing, every term in the sum is strictly increasing in $\pi_j(\mathbf{a})$ for $j \in \mathcal{R}$. For $j \notin \mathcal{R}$ with $j \neq I(\mathbf{a},\mathbf{b})$, $\pi_j(\mathbf{a})$ does not appear in the formula at all. For $j = I(\mathbf{a},\mathbf{b})$ (when $a_{I(\mathbf{a},\mathbf{b})} \neq b_{I(\mathbf{a},\mathbf{b})}$, so $I \notin \mathcal{R}$), each term is strictly decreasing in $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$ because ϕ is. Thus, Fermi imitation dynamics satisfies condition (2). □

3.3 Purely Extrinsic population dynamics in Two-action games

We now consider purely extrinsic population dynamics in games with two actions, C and D. We first define a further classification of population dynamic:

Definition 2 (ND-monotone population dynamic)**.** *Let T be the transition matrix for a population dynamic on a game with two actions, $\{C, D\}$, and equip the state space $S = \{C, D\}^N$ with the componentwise partial ordering with $D < C$. Let T^{nd} denote T with all payoffs set to a common value (we denote by $v_{\mathbf{a}}$ the common value in state $\mathbf{a}$). We say that T defines an ND-monotone population dynamic if:*

1. *$T^{nd}_{\mathbf{ab}}$ is independent of $v_{\mathbf{a}}$.*
2. *T^{nd} For every pair of states $\mathbf{a} \leq \mathbf{a}'$, there exist random variables $\mathbf{X}, \mathbf{X}'$ on a common probability space, with $\mathbf{X} \sim T^{\mathrm{nd}}(\mathbf{a}, \cdot)$ and $\mathbf{X}' \sim T^{\mathrm{nd}}(\mathbf{a}', \cdot)$, such that $\mathbf{X} \leq \mathbf{X}'$ almost surely.*

ND-monotonicity captures configuration neutrality: under the same neutral payoffs, starting from a more cooperative profile cannot make the next state any less cooperative, and the chosen common value does not impact this neutrality. In a process which is not ND-monotone, there could be a configuration-dependent pull toward a given action type at neutral payoffs, which is an inherant bias rather than truly neutral behaviour. It can be shown that this definition holds for both the Moran process, and Fermi imitation dynamics:

Theorem 3. *The Moran process and Fermi imitation dynamics are ND-monotone under action-invariant mutation.*

Proof. We begin with the Moran process. Set $\pi_i(\mathbf{a}) = v$ for every player i gives $f_i(\mathbf{a}) = 1 + \epsilon v$ for every i, whence $T_{\mathbf{ab}} = \frac{|\mathcal{R}|(1+\epsilon v)}{N \cdot N(1+\epsilon v)}(1 - \sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N} = \frac{|\mathcal{R}|}{N^2}(1 - \sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$, which is independent of v. Therefore, condition (1) holds

Under T^{nd}, as defined above, the Moran process samples a focal player i uniformly, a role model k uniformly from the population (since all fitnesses are equal), and a mutation outcome m from the action-invariant mutation distribution; player i then adopts a_k, or m if a mutation occurs. Couple the chains started at $\mathbf{a} \leq \mathbf{a}'$ on the common probability space generated by (i, k, m), applying the same draw to both. Coordinate i in the successor of $\mathbf{a}$ is a_k (or m); coordinate i in the successor of $\mathbf{a}'$ is a'_k (or the same m). Since $\mathbf{a} \leq \mathbf{a}'$ gives $a_k \leq a'_k$, and every other coordinate j is left unchanged with $a_j \leq a'_j$, the two successors $\mathbf{X}$ and $\mathbf{X}'$ satisfy $\mathbf{X} \leq \mathbf{X}'$ pointwise on this probability space, so condition (2) holds.

For Fermi imitation dynamics, if $\pi_i(\mathbf{a}) = v$ for every player i, then every payoff difference appearing in the Fermi formula is zero, so $\phi(\pi_I(\mathbf{a}) - \pi_j(\mathbf{a})) = \phi(0) = \frac{1}{2}$ in each summand, and $T_{\mathbf{ab}} = \frac{|\mathcal{R}|}{2N(N-1)}(1 - \sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$, independent of v. Therefore, condition (1) holds.

Under T^{nd}, the dynamic samples a focal player i uniformly, a role model k uniformly from the remaining $N-1$ players, an independent Bernoulli outcome with success probability $\phi(0) = \frac{1}{2}$, and a mutation outcome m from the action-invariant mutation distribution; player i adopts a_k on a success (or m if a mutation occurs) and otherwise keeps a_i. Couple the chains started at $\mathbf{a} \leq \mathbf{a}'$ on the common probability space generated by $(i, k, \mathrm{Bernoulli}, m)$, applying the same draw to both. On a mutation step, both successors set coordinate i to the same m. On a successful imitation step, they set coordinate i to a_k and a'_k respectively, with $a_k \leq a'_k$. On a no-update step, coordinate i retains $a_i \leq a'_i$. Every other coordinate is unchanged, so the two successors $\mathbf{X}$ and $\mathbf{X}'$ satisfy $\mathbf{X} \leq \mathbf{X}'$ pointwise on this probability space, and condition (2) holds. ☐

We will now show that a there is a theoretic limit on the abundance of cooperators in any ND-monotone purely extrinsic population dynamic. Before stating the theorem we record a comparison result for Markov chains on partially ordered spaces. The result is classical and we claim no novelty; we include a short proof for the reader's convenience, since its use in the proof of the theorem below may not be familiar to the reader.

Lemma 1. *Let P and Q be irreducible, aperiodic transition matrices on a finite partially ordered set $(S, \leq)$, with stationary distributions π^P and π^Q. Suppose that*

(i) $P(\mathbf{a}, \cdot) \preceq_{st} Q(\mathbf{a}, \cdot)$ for every $\mathbf{a} \in S$; and

(ii) Q is stochastically monotone: $\mathbf{a} \leq \mathbf{a}'$ implies $Q(\mathbf{a}, \cdot) \preceq_{st} Q(\mathbf{a}', \cdot)$.

Then $\pi^P \preceq_{st} \pi^Q$.

Proof. Kamae, Krengel, and O'Brien [28] prove that whenever two probability measures μ_1, μ_2 on a partially ordered Polish space satisfy $\mu_1 \preceq_{st} \mu_2$, there exist random variables $X_1 \sim \mu_1$ and $X_2 \sim \mu_2$ defined on a common probability space with $X_1 \leq X_2$ almost surely (a *monotone coupling*).

Using this, we construct a coupling of the two chains by induction on t. Start with any $X_0^P, X_0^Q \in S$ satisfying $X_0^P \leq X_0^Q$. Given $X_t^P \leq X_t^Q$, hypothesis (ii) gives $Q(X_t^P, \cdot) \preceq_{st} Q(X_t^Q, \cdot)$, which combined with (i) at $\mathbf{a} = X_t^P$ yields $P(X_t^P, \cdot) \preceq_{st} Q(X_t^Q, \cdot)$. Applying Kamae–Krengel–O'Brien to this pair of measures produces (X_{t+1}^P, X_{t+1}^Q) with the required marginals and $X_{t+1}^P \leq X_{t+1}^Q$ almost surely. By induction, $X_t^P \leq X_t^Q$ for every t. Since both chains are ergodic, their marginal laws converge to π^P and π^Q, and passing to the limit gives $\pi^P \preceq_{st} \pi^Q$. ☐

3.3.1 The Ceiling of Cooperation

We now consider the ceiling on the average abundance of cooperatorss in purely extrinsic population dynamics.

Theorem 4 (Extrinsic Ceiling)**.** *Let T define an ND-monotone purely extrinsic process on an N-player game with 2 actions, C and D, such that in any given state $\mathbf{a}$, $\min_{j:a_j=D} f_j(\mathbf{a}) > \max_{i:a_i=C} f_i(\mathbf{a})$ (that is, in any given state, any defector has greater fitness than any cooperator), under action-invariant mutation.*

Then the mean number of cooperators in the steady state of T, denoted p_C, satisfies $p_C \leq \frac{1}{2}$

Proof. Equip $S = \{C, D\}^N$ with the componentwise partial order, declaring $D < C$. For a transition matrix P, write $P(\mathbf{a}, \cdot) \preceq_{st} P(\mathbf{a}', \cdot)$ to mean that the one-step distribution from $\mathbf{a}$ under P is stochastically dominated by the one-step distribution from $\mathbf{a}'$ under P in this order. Let $T^{\mathrm{nd}}_{\mathbf{ab}}$ denote the value of $T_{\mathbf{ab}}$ evaluated at $\pi_i(\mathbf{a}) = 0$ for every i; by condition (4), $T^{\mathrm{nd}}_{\mathbf{ab}}$ is equally the value of $T_{\mathbf{ab}}$ under any other choice of common payoff.

Step 1: $T(\mathbf{a}, \cdot) \preceq_{st} T^{\mathrm{nd}}(\mathbf{a}, \cdot)$ for every $\mathbf{a} \in S$.

Set $v(\mathbf{a}) := \min_{a_j = D} \pi_j(\mathbf{a})$. In every mixed state $\mathbf{a}$, the payoff-dominance hypothesis implies $\pi_i(\mathbf{a}) < v(\mathbf{a})$ for every cooperator i and $\pi_j(\mathbf{a}) \geq v(\mathbf{a})$ for every defector j. Consider a C-gaining transition $\mathbf{a} \to \mathbf{b}$, so $b_{I(\mathbf{a},\mathbf{b})} = C$ and $\mathcal{R}$ is the set of current cooperators. Replacing every $\pi_i(\mathbf{a})$ with $v(\mathbf{a})$ raises all role-model payoffs and lowers all non-role-model payoffs, so by condition (2),

$$T_{\mathbf{ab}} \;\leq\; T_{\mathbf{ab}}\big|_{\pi_i(\mathbf{a}) = v(\mathbf{a}) \text{ for all } i}$$

with strict inequality whenever $\mathcal{R} \neq \emptyset$. By condition (4), the right-hand side equals $T^{\mathrm{nd}}_{\mathbf{ab}}$. The analogous argument with reversed inequality applies to D-gaining transitions. Hence $T(\mathbf{a}, \cdot) \preceq_{st} T^{\mathrm{nd}}(\mathbf{a}, \cdot)$.

Step 2: T^{nd} is invariant under the $C \leftrightarrow D$ label swap.

Let $\sigma : S \to S$ be the map that swaps every C and D. By conditions (3) and (4), with common payoff 0, the only features of $(\mathbf{a}, \mathbf{b})$ that enter $T^{\mathrm{nd}}_{\mathbf{ab}}$ are the mutation matrix and the set $\mathcal{R}$. Under action-invariant mutation the mutation matrix is σ-invariant; and $|\mathcal{R}|$ is σ-invariant because the set of players currently playing $b_{I(\mathbf{a},\mathbf{b})}$ in $\mathbf{a}$ has the same size as the set of players currently playing $\sigma(b_{I(\mathbf{a},\mathbf{b})})$ in $\sigma(\mathbf{a})$. Hence $T^{\mathrm{nd}}_{\mathbf{ab}} = T^{\mathrm{nd}}_{\sigma(\mathbf{a})\sigma(\mathbf{b})}$ for every transition, so the stationary distribution of T^{nd} is σ-invariant, giving $p_C^{(T^{\mathrm{nd}})} = p_D^{(T^{\mathrm{nd}})} = \frac{1}{2}$ [41].

Step 3: T^{nd} is stochastically monotone.

Condition (5) provides, for every $\mathbf{a} \leq \mathbf{a}'$, a coupling $(\mathbf{X}, \mathbf{X}')$ on a common probability space with $\mathbf{X} \sim T^{\mathrm{nd}}(\mathbf{a}, \cdot)$, $\mathbf{X}' \sim T^{\mathrm{nd}}(\mathbf{a}', \cdot)$, and $\mathbf{X} \leq \mathbf{X}'$ almost surely. For any upper set $U \subseteq S$, $\{\mathbf{X} \in U\} \subseteq \{\mathbf{X}' \in U\}$, so $T^{\mathrm{nd}}(\mathbf{a}, U) \leq T^{\mathrm{nd}}(\mathbf{a}', U)$, giving $T^{\mathrm{nd}}(\mathbf{a}, \cdot) \preceq_{st} T^{\mathrm{nd}}(\mathbf{a}', \cdot)$.

Applying Lemma 1 to $P = T$ and $Q = T^{\mathrm{nd}}$ (which satisfy (i) by Step 1 and (ii) by Step 3) yields $\pi^T \preceq_{st} \pi^{T^{\mathrm{nd}}}$. Since n_C is monotone in the componentwise order,

$$p_C^{(T)} \;=\; \tfrac{1}{N}\,\mathbb{E}_T[n_C] \;\leq\; \tfrac{1}{N}\,\mathbb{E}_{T^{\mathrm{nd}}}[n_C] \;=\; p_C^{(T^{\mathrm{nd}})} \;=\; \tfrac{1}{2}.$$

□

Our result is comparable to that of [3], which showed that for the game defined by 1 and a wide range of evolutionary processes with mutation, the condition $a(N-2) + bN > cN + d(N-2)$ is crucial for a strategy A to dominate strategy B in pairwise comparison processes at any mutation rate. This builds upon the work of [29, 43], generalising it to arbitrary mutation and selection intensity. We now show that for ND-monotone purely extrinsic population dynamics, pointwise domination of defection over cooperation is sufficient for defection to dominate cooperation in a heterogeneous population. This extends the understanding of when defection dominates cooperation to any population, game, and population dynamic such that the above conditions are satisfied.

Crucially, we show that even if a player may increase their payoff by a switch $D \to C$ at all times, pointwise dominance of D will still cause the process to favour D. In the words of a 2x2 symmetric game as discussed in [3, 29, 43], pointwise domination corresponds to a system where:

$$\frac{j-1}{N-1}a + \frac{N-j}{N-1}b > \frac{j}{N-1}c + \frac{N-j-1}{N-1}d \tag{5}$$

where $j < N$ is the number of D players. In 2x2 pairwise games, it is not possible to formulate a system where 5 is satisfied and yet in a given state a change $D \to C$ improves a player's fitness, as the following would need to be satisfied:

$$\frac{j-1}{N-1}a + \frac{N-j}{N-1}b < \frac{j+1-1}{N-1} + \frac{N-(j+1)}{N-1} = \frac{j}{N-1} + \frac{N-j-1}{N-1} \tag{6}$$

which contradicts 5.

An example of this is the public goods game with $r > N$. This satisfies both pointwise dominance and a fitness increase for any player under the $D \to C$ swap, and so we can see that it cannot be described as pairwise interactions in the manner of [3, 29, 43]. In such a game, there is no temptation to free ride; players will obtain a greater payoff by cooperating regardless of the population structure. The case of such a public goods game is considered in figure 3 of [20], where a subset of a homogeneous population play a public goods game at each time step. The following result extends previous work to the general set of games where cooperators outperform defectors at a given times, as well as to heterogeneous populations. Our result gives the ceiling on the abundance of cooperation in ND-monotone purely extrinsic population dynamics.

This shows that a player's chosen strategy does not necessarily favour the action type which maximises their payoff in *either* the one-step or the long-run strategy distribution, instead favouring the strategy which is pointwise dominant over its peers. Even if there is no temptation to free-ride, players will still choose to defect based on the pointwise dominance of defectors in the state.

3.3.2 Purely extrinsic population dynamics in the Public Goods Game

To illustrate for the results of section 3, we carry out a large parameter sweep across parameters of a public goods game [51, 25, 4, 24, 23]. In such a game, players choose to contribute some amount α_i. The total contribution is multiplied by some factor $r > 0$ and split between all players. We consider heterogeneous contributions and a homogeneous return.

The parameter sweep is done to the highest standards of reproducibility [62, 47, 63, 53, 56] and all data has been archived at [?]and is a further contribution of the work. In order to perform this sweep, the authors have developed an installable software library `ludics` (written in python). The archive of `ludics` is at [17] and the documentation can be found at [16]. The full development of the software can be found at [15]. The parameters considered, as well as their meanings in reference to the public goods game are considered in table 3.

Parameter	Meaning	Minimum	Maximum	Number of Values per N
N	Number of players	3	8	N/A
r	Return on investment	$\frac{1}{2}$	$\frac{3N}{2}$	10
α_i	Contribution of player i	N/A	N/A	N/A
M	Maximum total contribution	N	$4N$	10
μ	Mutation value	0.001	0.1	4
ϵ	Selection intensity	0	$0.99 \cdot \max_i \alpha_i$	100
β	Choice intensity	0	2	10
A	Aspiration	1	M	100

Table 3: The set of measured parameters and boundry values. Some parameters are only applicable to certain processes. α_i is entirely controlled by the value M, which controls the maximum total contribution. The distribution of wealth remains the same for a given N, however as M increases, so too does the size of each player's contribution α_i. The value A is the *aspiration* considered in aspiration dynamics, in section 4. We consider a homogeneous return r. Values of ϵ and A depend on M rather than N, and as such there are a large number of values per N for these parameters.

Our results for the purely extrinsic dynamics are shown in Figure 2, and we explore Theorem 4 numerically in the context of the 8-player public goods game. All other values of N are given as supplimentary information. In panels (a) through (e), we can see this limit as a hard ceiling on the value of p_C which no parameter set in our data is able to exceed. This includes parameter combinations which have high values of $r > N$, and so we see that despite full cooperation being the Nash equilibrium of the one shot game, and increasing the payoff of players in the one-step distribution, the pointwise dominance of defection's fitness over that of cooperation leads to defection becoming the dominant strategy in a repeated game according to a purely extrinsic population dynamic. This not only leads to a worse average payoff for all players in the population in the steady state, but also means that when $r > N$, worsening moves occur with a higher probability than changes of action types which improve a player's payoff. In other words, even when a switch $D \to C$ would always improve a player's payoff, the player will still prefer to remain a defector under a purely extrinsic population dynamic, due to the greater fitness of a defector in a given state.

In Fermi imitation dynamics, we can also note the process' invariance under r. Figures 2a and 2d show an identical distribution of p_C for all values of r. The latter of these two figures also give that an increase in β will have a negative effect upon p_C, as the value $\Delta(\pi)$ will always be negative for a $C \to D$ transition, and positive for a $D \to C$ transition. As the process with $\beta = 0$ gives us neutral drift, we see our result in practice, as moving towards a less rational process increases p_C in all empirical cases, up to the neutral-drift baseline. The notion of "rationality" in a purely extrinsic process is not directly linked to increasing a player's payoff. p_C in the Moran process increases with the value of r, however once again we see that it has a hard limit at $p_C = \frac{1}{2}$, as a given defector is always more likely to be chosen for reproduction than a given cooperator.

One interesting point is the invariance of the Moran process under the size of each player's contribution. The probability of selecting a defector in a state with q defectors is given by:

$$\frac{qr\sum_{j=1}^{N}\alpha_j}{Nr\sum_{j=1}^{N}\alpha_j-\sum_{\mathbf{a}_l=C}\alpha_l}=\frac{qr\sum_{j=1}^{N}\alpha_j}{(Nr)\sum_{j=1}^{N}\alpha_j-\sum_{j=1}^{N}\alpha_j}=\frac{qr\sum_{j=1}^{N}\alpha_j}{(Nr-1)\sum_{j=1}^{N}\alpha_j}=\frac{qr}{Nr-1} \tag{7}$$

The analogous result for choosing a defector is given by $1-\frac{qr}{Nr-1}$, which is once again invariant under α_i. Therefore, as Fermi imitation dynamics is invariant under the return parameter r, the Moran process is invariant under the contribution parameter α_i. We can observe this effect in figures 2f and 2g.

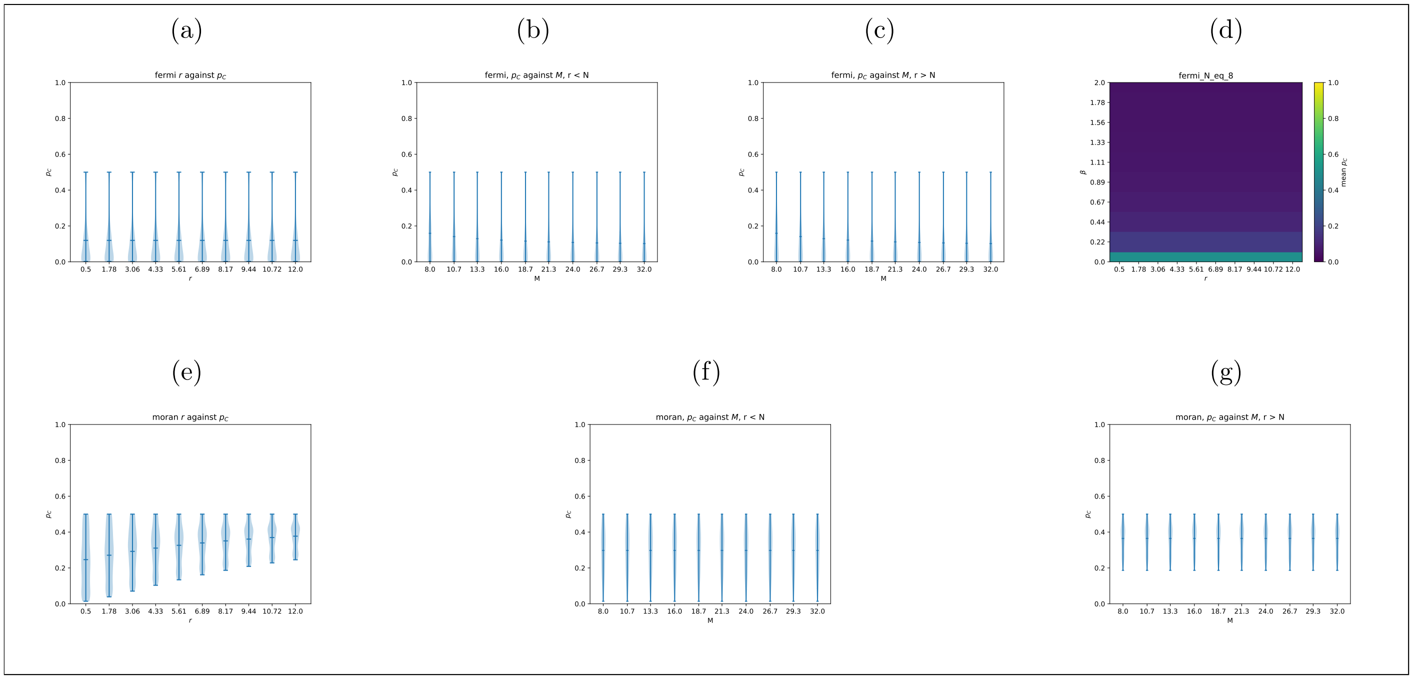


Figure 2: (a): The distribution of p_C in Fermi imitation dynamics for different values of r. Since Fermi imitation dynamics is invariant under the value of r, this distribution does not change, and for all values of r it admits a maximum value of p_C below the neutral-drift baseline $p_C = \frac{1}{2}$. (b) & (c): The distribution of p_C for Fermi imitation dynamics for different sizes of contribution. The greater the value of M, the greater the size of each player's contribution. (b) shows when $r < N$, and (c) shows $r > N$. (d): A heatmap of mean(p_C) for values of r and β for Fermi imitation dynamics. We see that there is a negative correlation between β and p_C. (e - g): The equivalent of (a - c) for the Moran process. (f): The distribution of difference in p_C between Fermi imitation dynamics and the Moran process for different combinations of shared parameters.

4 Intrinsic Population Dynamics

We have seen how purely extrinsic population dynamics only compare a player's payoffs to the payoff in the current state. We now see a classification of population dynamic where players instead compare their payoff to some attribute or payoff other than that of other players. We define a *purely intrinsic population dynamic* as follows:

Definition 3 (Purely intrinsic population dynamic)**.** *A population dynamic defined by a transition matrix T is said to be* purely intrinsic *if it satisfies the following conditions:*

1. *$T_{\mathbf{ab}}$ is a function of the payoffs of no player other than $I(\mathbf{ab})$.*
2. *$T_{\mathbf{ab}}$ is decreasing in the payoff $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$*
3. *$T_{\mathbf{ab}}$ contains no term dependent on $a_{I(\mathbf{a},\mathbf{b})}$ or $b_{I(\mathbf{a},\mathbf{b})}$, except in the payoff function $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$, mutation parameters μ_{ij}, and in the definition of the set $\mathcal{R}$.*

We now consider two purely intrinsic population dynamics - aspiration dynamics, which compares a player's payoff to their individual baseline aspiration, and introspection dynamics, which compares a player's payoff to their counterfactual payoff when they change action type.

4.1 Aspiration Dynamics

2b

Under aspiration dynamics, players choose to change action based on how far away their current payoff is from a target payoff. It follows the algorithm:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy
2. This player changes strategy with a probability $\phi(\pi_i(\mathbf{a}) - A_i)$

Here, A_i is player i's *aspiration* - the payoff which they desire to make from a given game. A player will accept a new strategy with a higher than random probability if their current payoff is lower than their aspired payoff. This dynamic is traditionally paired with two action games [12, 11, 66], though it can be defined for games with 3 or more actions [65]. Aspiration dynamics has also often being combined with imitation dynamics to form a new population dynamic [33, 46, 5, 58]. We the original form of aspiration dynamics defined in [12], with a homogeneous aspiration value A.

The transition matrix for aspiration dynamics is given by:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N}\,\phi(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - A_{I(\mathbf{a},\mathbf{b})})\left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{8}$$

The traditional two-action model of aspiration dynamics is very specific in which games it can be applied to. When aspiration is considered for three or more action types [65], a player chooses which action to consider with equal probability. Thus, players in 3-action games do not choose their strategy in the next state based on the fitness of said strategy, but instead update hrough random chance in order to reach their aspiration. Players may accept worsening moves with a high probability.

This population dynamic spends a large amount of time in states where most players achieve their aspired payoffs. In this case, players will have a lower probability of switching strategy, compared to a state where players are underperforming in comparison to their target A. Thus, cooperation emerges more frequently in aspiration dynamics under larger values of A, as more cooperators are required in order for all players to reach their aspired payoff. At a lower value of A, the reverse is true - fewer cooperators are required for all players to reach their target payoff. We now show that Aspiration dynamics is a purely intrinsic population dynamic:

Theorem 5 (Aspiration dynamics is a purely intrinsic population dynamic)**.**

Proof. It is clear from the definition of $T_{\mathbf{ab}}$ that aspiration dynamics satisfies conditions (1) and (3). It remains to show that aspiration dynamics is decreasing in the payoff $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$. Since $\phi(x)$ is decreasing, and $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$ appears only in the value $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - A_i$. Therefore, setting $x = \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$, we get that ϕ is decreasing in $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$, and therefore so is $T_{\mathbf{ab}}$. □

4.2 Introspection Dynamics

Unlike the previous population dynamics we have seen, introspection dynamics [9, 10] allows players to consider the payoff they will have if they change action. Under this population dynamic, players compare their current payoff with their counterfactual payoff upon changing action type. The process is give by the algorithm:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy
2. Player i randomly chooses another possible strategy, a_{il}, from the set of all possible strategies.
3. The chosen player replaces their strategy with probability $\phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})})$, where $\Delta\pi_{I(\mathbf{a},\mathbf{b})} = \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{b})$ is the difference between the fitness of the player's current strategy and the possible fitness of the new strategy.

Introspection dynamics is well suited to a heterogeneous population, as players are able to consider whether a strategy will work for them, rather than for another player. As players check the payoff of a strategy in the next state, they do not accept worsening moves with as high of a probability, as the value $\phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})})$ is greater than $\frac{1}{2}$ if and only if the change in a player's strategy increases their payoff. The transition matrix is given by:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N(k-1)}\,\phi(\Delta(\pi_{I(\mathbf{a},\mathbf{b})}))\left(1-\sum_{j=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),j}\right)+\frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b}\in \mathrm{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b}\notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a}\neq\mathbf{b}, \\ 1-\sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a}=\mathbf{b}. \end{cases} \tag{9}$$

Introspection dynamics is aligned with classic static game theory, where a well informed and rational actor maximises their own payoff. We can contrast this with purely extrinsic population dynamics, which are the traditional method of modelling populations in evolutionary games, where actors are not typically well-informed and choose their next strategy according to fitnesses in the current population, without knowledge of the impact of their transition on their fitness.

Definition 4 (Introspection dynamics is a purely intrinsic population dynamic)**.**

Proof. It is once again clear that introspection dynamics satisfies conditions (1) and (3). For condition (2), consider that $\phi(x)$ is decreasing. Setting $x = \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{b})$, we have our result. ☐

Unlike purely extrinsic population dynamics, introspection dynamics does not necessarily cancel out certain parameters of games, unless said parameter does not affect the difference in a player's payoff when playing different action types. For example, if we consider the public goods game, we can see that for introspection dynamics, $\Delta(\pi_{I(\mathbf{a},\mathbf{b})}) = \alpha_i(\frac{r}{N}-1)$, which considers both α_i and r.

4.3 Intrinsic population dynamics in a public goods game

In Figure 3, we see that intrinsic population dynamics are able to exceed the ceiling on p_C which is imposed on purely extrinsic population dynamics. In the case of aspiration dynamics, p_C does not reach excessively low values as in the other processes we have discussed. This is because for all players to have a payoff $\pi_i > 0$, at least one player must be cooperating at all times. If no players cooperate, then no player will reach their target payoff threshold, and players will switch strategy with a high probability. The ceiling is not exceeded for $r < N$, as showing in figure 3a and noted in [12], but -is exceeded at the threshold $r > N$. In this case, $\pi_i(\mathbf{a}) - A$ is greater when cooperating than defecting in a given state $\mathbf{a}_{-i}$ for all players, and thus the transition $D \to C$ will always occur with a higher probability than the reverse. For intermediate values of r, aspiration dynamics can achieve its lowest p_C, as with a low value of A, players are able to achieve their aspiration by free-riding in a state with fewer defectors. Figure 3d shows that the mean value of p_C remains fairly steady for the values of r and β. The aspiration level is the biggest indicator of cooperation in the population in a linear public goods game where a player's fitness is given by their expected return in a 2-player game. It has been shown previously that for a high aspiration level, cooperation emerges at a level near the neutral-drift baseline, but for a low aspiration level defectors dominate the population [12].

As the heterogeneous public goods game is an additive game [**?**], we can show that many of the analytical results of [18] about the structure of introspection dynamics in a public goods game hold empirically. Each player i cooperates with probability $p_i = \phi_i(\alpha_i(1-\frac{r}{N}))(1-2\mu)+\mu$, and p_C is given as a product measure [**?**]. Thus, we have that as the value $\Delta(\pi_{I(\mathbf{a},\mathbf{b})}) = \alpha_i(\frac{r}{N}-1)$ is positive for $r > N$ and negative for $r < N$, there is a sharp turning point at $r = N$ where introspection dynamics begins to favour cooperation, as shown in figure 3e. This gives an exact parameter value at which the ceiling of p_C is overcome.

The figures 3f and 3g show the result that $\frac{\partial p_C}{\partial \alpha_i} > 0$ if and only if $r > N$, as the value $\alpha_i(\frac{r}{N}-1)$ becomes greater in this case, but becomes more negative if $r < N$. A greater contribution will increase a player's payoff if multiplied by a large r, but for a small r a larger contribution results in a greater loss to the player.

The same conditions also apply to the effect of β, as shown in 3h. This is because $\phi_i(\Delta(\pi_{I(\mathbf{a},\mathbf{b})}))$ is decreasing. Therefore, as β scales $\Delta(\pi_{I(\mathbf{a},\mathbf{b})})$, when $\Delta(\pi_{I(\mathbf{a},\mathbf{b})})$ is negative, $\frac{\partial p_C}{\partial \alpha_i} > 0$, and the inequality is reversed if the opposite is true. We acquire $\frac{\partial p_C}{\partial \alpha_i} < 0$ at exactly the value $r > N$, and thus we have our result. These results are proven in [18], but have been shown empirically for our large data sweep in figure 3.

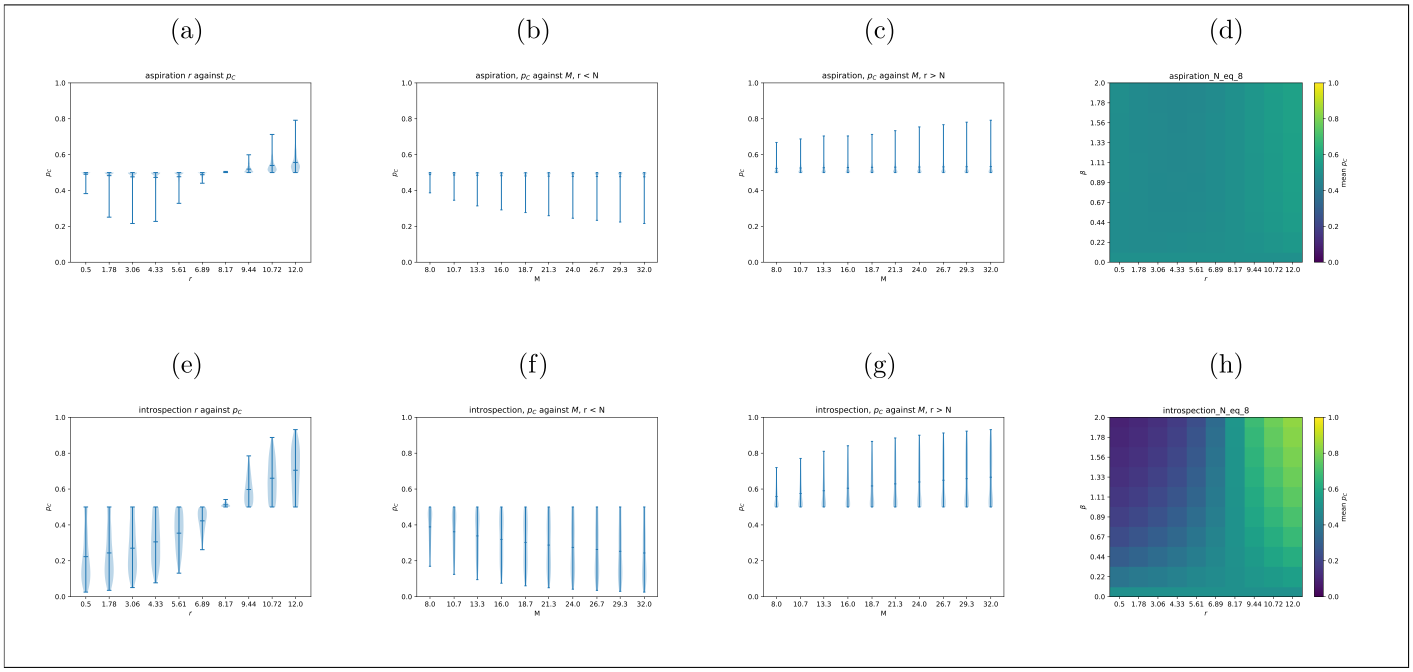


Figure 3: (a): The distribution of p_C in aspiration dynamics. For medium values of r, players are able to achieve their aspired fitness with fewer cooperators, reducing the value of p_C. However, when $r > N$, players obtain a higher payoff in a state $\mathbf{a}$ where they cooperate than in the state $\mathbf{a}_{-i}$ where they defect. Consequently, $p_C > \frac{1}{2}$ in this regime.
(b)–(c): The distribution of p_C for aspiration dynamics for different magnitudes of contribution. Higher deciles of range(α_i) correspond to larger player contributions. Panel (b) shows the case $r < N$, while panel (c) shows $r > N$.
(d): A heatmap of mean(p_C) for values of r and β under aspiration dynamics.
(e)–(h): The equivalent results for introspection dynamics.

4.4 The weaknesses of purely intrinsic population dynamics

While purely intrinsic population dynamics are able to exceed the ceiling placed on p_C by purely extrinsic population dynamics, they are not without their own inherant limitations. While the property of introspection dynamics under additive games shown in **??** is useful for obtaining analytical results, it shows that introspection dynamics does not take into account the remainder of the state's actions. Players pay no attention to how successful a strategy is in the population, and instead consider strategies at random, accepting them with a probability which often does not depend on the actions of their co-players (that is, the state $\mathbf{a}_{-i}$).

Aspiration dynamics is not invariant under the actions of other players (assuming the actions of other players affect a given player's payoff, as in the public goods game), however it still does not consider the fitness of other strategies when accepting them. A player may achieve a payoff beneath their target by defecting, and so will change strategy with a probability greater than $\frac{1}{2}$. However, the adopted strategy is not accepted according to it's payoff, either in the current state or counterfactual, and so the accepted strategy may be a worsening move with a high probability. Additionally, a player may be satisfied too easily, not changing strategy in favour of a better one as they have already reached their aspiration.

As a result of these shortcomings, and the ceiling on p_C in purely extrinsic population dynamics, we introduce a new population dynamic which can overcome the drawbacks of the previous four.

5 Introspective Imitation Dynamics

We now introduce a novel population dynamic *introspective imitation dynamics.* This process combines the purely intrinsic decision step of introspection dynamics with the purely extrinsic fitness-proportional selection of the Moran process, such that each individual takes into account both the payoff of strategies in the current state, and the counterfactual payoff obtained when accepting a new strategy. This models both extrinsic learning and intrinsic learning, and is defined neither as purely extrinsic or purely intrinsic. We define introspective imitation dynamics by the following algorithm:

1. A focal player is chosen at random to reconsider their strategy
2. Another player a_j in the population is chosen with a probability proportional to their fitness within the population to have their strategy considered, as in the Moran process.
3. The focal player changes their strategy with probability $\phi(\Delta\pi_i)$, as in introspection dynamics

Recall that $f_i(\mathbf{a}) = 1 + \epsilon \cdot \pi_i(\mathbf{a})$. We define the transition matrix T of an introspective imitation process as follows:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N}\dfrac{\sum_{j:\, a_j = b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{\sum_l f_l(\mathbf{a})}\,\phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})})\left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{10}$$

This models a behaviour where individuals learn from each other, but do not blindly trust that the strategy will work for themselves solely because it worked for another player in the population. Instead, players look to the payoff of others in the population to see what action they *could* play, but judge based on their own counterfactual payoff to decide what action they *should* play. This means that introspective imitation dynamics is well-suited to a heterogeneous population, while not ignoring the sucess of strategies in a given state.

Introspective imitation dynamics does not break down into individual Markov chains like introspection dynamics due to the extrinsic step, and is defined for any number of strategies. It considers each strategy proportional to both its payoff within the state and the counterfactual payoff which the focal player is able to obtain by adopting the new action type, and as such it can take both a choice intensity β and a selection intensity ϵ. This gives introspective imitation dynamics the ability to emphasise either the extrinsic or intrinsic step of the dynamic, which is suited to game theoretic models of real-world systems where different individuals make decisions based on different factors. We can also describe introspective imitation dynamics as a meeting point of classic static game theory with traditional evolutionary game theory. The fitness-proportional selection step being identical to that of the Moran process, and the intrinsic decision step being analogous to a well-informed rational actor attempting to maximise their own payoff.

We now show introspective imitation dynamics in a public goods game and compare it's results to that of the previous four population dynamics.

5.1 Introspective Imitation Dynamics in the Public Goods Game

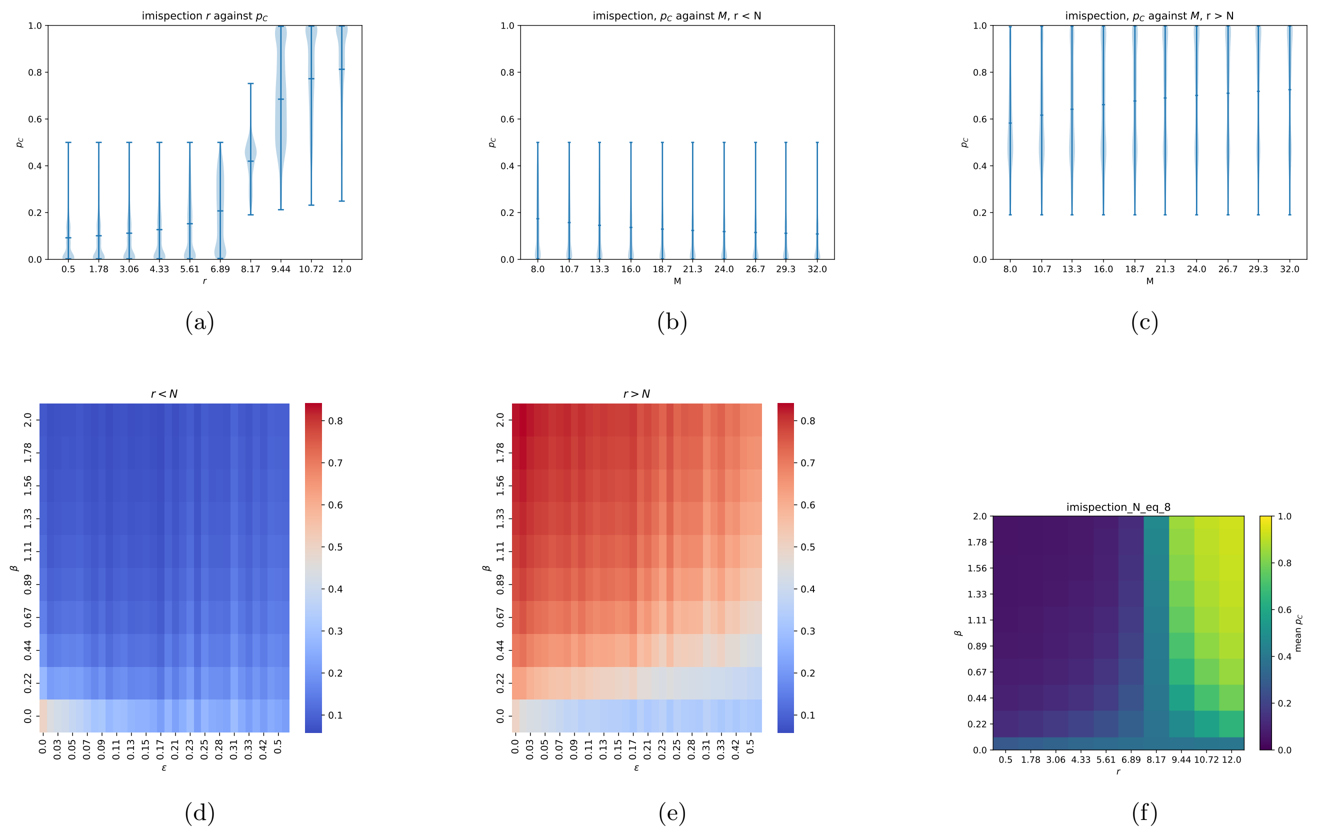


Figure 4: (a): the distribution of p_C in introspective imitation dynamics for various values of r.
(b), (c): the distribution of p_C in introspective imitation dynamics for various values of M. (b) shows $r < N$, (c) shows $r > N$
(d),(e): Heatmaps of the the mean(p_C) for the data across pairs of selection and choice intensity, for $r > N$ and $r < N$ respectively.
(f): A heatmap of mean(p_C) in introspective imitation dynamics for varying r and β. We see a threshold for the effect of a stronger choice intensity at $r = N$.

We see in Figure 4a that in the same manner as the purely intrinsic population dynamics, introspective imitation dynamics is able to exceed the $p_C = \frac{1}{2}$ ceiling which is imposed by purely extrinsic population dynamics at the threshold $r = N$. Despite the extrinsic step in the dynamic, the intrinsic step is sufficient to enable the dynamic to admit high values of p_C under the right parameters. However, unlike the purely intrinsic population dynamics which we have seen, the $r = N$ threshold does not *guarantee* the exceeding of the cooperation ceiling. For certain values of ϵ and β, the extrinsic step supersedes the intrinsic step, and surpresses the probability of a $D \to C$ transition, as shown in figures 4d and 4e, reducing the overall p_C. However, for a given pair (ϵ, β) with $\beta > 0$, there is no result which guarentees a $p_C < \frac{1}{2}$; where the value falls with regards to this threshold is reliant upon r.

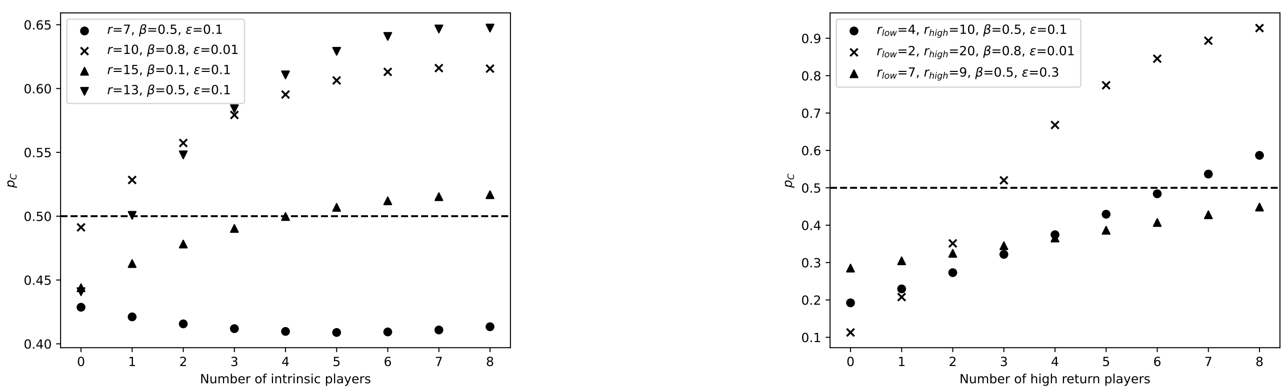
Introspective imitation dynamics also satisfies the same threshold for $\frac{\partial p_C}{\partial \beta}$ as introspection dynamics, giving us that when $r > N$, $\frac{\partial p_C}{\partial \beta} > 0$, but the opposite is true when $r < N$. Figure 4f shows this effect, and it is clear that as β only appears in the intrinsic step, it's effect would have the same effect as in the purely intrinsic dynamic. For this same reason, the $r > N$ requirement for $\frac{\partial p_C}{\partial \alpha_i} > 0$ is also equivalent to that of introspection dynamics, with the inequality reversed for $r < N$. The purely extrinsic step of the process cancels out α_i, as seen in equation 7. We can see this effect in figures **??** and **??**.

When $r < N$, the intrinsic step of the dynamic surpresses cooperation compared to a Moran process, taking a value $\phi < \frac{1}{2}$ for a $D \to C$ transition, and $\phi > \frac{1}{2}$ for a $C \to D$ transition.

6 Conclusion

We have introduced and analysed a framework for heterogeneous evolutionary games under five distinct population dynamics in three categories, as shown in Figure 2. We have defined the categorisation of a *purely extrinsic population dynamic*, and given rigorous formulations of two widely used examples in the Moran process and Fermi imitation dynamics. We have shown that dynamics under this classification have a hard ceiling on the probability of cooperation p_C in social dilemmas where a defector *always* outperforms a cooperator in a given state. In such a game, purely extrinsic population dynamics are unable to exceed the neutral-drift baseline. We then give formulations of two purely intrinsic population dynamics. Aspiration dynamics in which players change strategy based on their performance in relation to a target payoff, and introspection dynamics in which players compare their current payoff with their counterfactual payoff which they could obtain by switching action type. We have also shown empirical results for the public goods game with heterogeneous contributions which confirm the extrinsic ceiling, as well as the ability for purely intrinsic population dynamics to exceed it. Finally, we have introduced a novel population dynamic in Introspective imitation dynamics with both an extrinsic and intrinsic step. Our empirical results confirm that this dynamic is able to overcome the ceiling of purely extrinsic population dynamics, but still allows players to take into account the performance of strategies within the current state.

Further work may look further into the effect of hybrid population dynamics, similar to the combination of Moran and Fermi processes in [35], where extrinsic and intrinsic population dynamics are played in a mixed population. Figure 5a shows three populations where the population dynamic is treated as a heterogeneous attribute, with players learning according to either the Moran process or introspective imitation dynamics. We see that under different parameters, a different number of players may possess an intrinsic learning method in order to exceed the neutral drift baseline p_C. The heterogeneity considered in this paper is also only in the player's contributions, however the threshold for introspective imitation dynamics occurs at the point $r = N$. Figure 5b shows that in introspective imitation dynamics with certain heterogeneous values of r_i, fewer players are required to receive a return $r > N$ in order to exceed the cooperation threshold. Additionally, we see in both Figures 5b and **??** that under certain values of ϵ and β, even a population which is entirely contributing a high value of r will be unable to exceed the p_C threshold, as the extrinsic step of the process contributes with too high intensity. The relationship between such values and the heterogeneity of r may be a focus of further study. Finally, we only consider the ceiling of p_C in two-action games for purely extrinsic population dynamics. It is entirely possible that there is a formulation of a $k-$action game such that a ceiling is posed on the abundance of a certain action. This may be a focus of future work, extending the limitation of purely extrinsic population dynamics to a larger set of games.



(a)

(b)

Figure 5: (a): p_C in various parameter combinations at $N = 8$, varying the number of players using an intrinsic decision process. 0 intrinsic players gives the p_C of the Moran process, whereas 8 gives us the p_C in introspective imitation dynamics. We take a linear population with a maximum contribution of 8. (b): p_C for a heterogeneous r in introspective imitation dynamics. 0 high return players corresponds to a population with homogeneous return r_{low}. 8 high return players corresponds to a population with a homogeneous return r_{high}.

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